

Exam Practice Problems:

Gradient Descent / RNN / CNN

Problem 1: Gradient Descent

Problem Statement

Given the function $f(x) = (x - 3)^2$, use gradient descent starting from $x_0 = 8$. The learning rate is $\alpha = 0.2$. Calculate the first two iterations to obtain x_1 and x_2 .

Solution

The derivative of the function is:

$$f'(x) = 2(x - 3)$$

First iteration:

$$x_1 = x_0 - \alpha f'(x_0) = 8 - 0.2 \cdot 2(8 - 3) = 8 - 2 = 6$$

Second iteration:

$$x_2 = x_1 - \alpha f'(x_1) = 6 - 0.2 \cdot 2(6 - 3) = 6 - 1.2 = 4.8$$

Answer

$$x_1 = 6, \quad x_2 = 4.8$$

Problem 2: RNN Sequential Computation

Problem Statement

A Recurrent Neural Network (RNN) updates hidden states using the formula:

$$h_t = \tanh(Wx_t + Uh_{t-1} + b)$$

Given: $W = 1$, $U = 1$, $b = 0$, initial state $h_0 = 0$, and input sequence $x = [1, 2]$. Calculate h_1 and h_2 (to four decimal places).

Solution

$$h_0 = 0$$

First step:

$$h_1 = \tanh(1 \cdot 1 + 1 \cdot 0) = \tanh(1) \approx 0.7616$$

Second step:

$$h_2 = \tanh(1 \cdot 2 + 1 \cdot 0.7616) = \tanh(2.7616) \approx 0.9920$$

Answer

$$h_1 \approx 0.7616, \quad h_2 \approx 0.9920$$

Problem 3: CNN Output Size Calculation

Problem Statement

Given an input image of size 32×32 , apply a convolution with kernel size 5×5 , padding = 0 (valid), and stride = 1. Calculate the output feature map size.

Solution

Use the formula:

$$\text{Output Size} = \left\lfloor \frac{N - F + 2P}{S} \right\rfloor + 1$$

Where: $N = 32$, $F = 5$, $P = 0$, $S = 1$

$$\text{Output Size} = \left\lfloor \frac{32 - 5 + 0}{1} \right\rfloor + 1 = 28$$

Answer

Output feature map size is 28×28